

2020; Benzell et al., 2021). The realization of general purpose technologies’ full potential requires extensive co-invention (Bresnahan and Trajtenberg, 1995; Bresnahan et al., 1996, 2002; Lipsey et al., 2005; Dixon et al., 2021), a costly and time-consuming process involving the discovery of new business procedures (David, 1990; Bresnahan, 1999; Frey, 2019; Brynjolfsson et al., 2021; Feigenbaum and Gross, 2021). Consequently, many studies of machine learning technologies focus on systems-level adoption, arguing that organizational systems may require redesign to effectively take advantage of novel machine learning advancements (Bresnahan, 2019; Agrawal et al., 2021; Goldfarb et al., 2023). Appropriately designed systems can yield considerable business value and improve firm performance (Rock, 2019; Babina et al., 2021; Zolas et al., 2021), with AI tools facilitating the discovery process (Cockburn et al., 2018; Cheng et al., 2022). By employing task-level information to assess whether LLMs fulfill the criteria of a general purpose technology, we seek to merge the two perspectives for understanding the technology-labor relationship.

We attempt to build on these diverse literature streams in several ways. Echoing (Felten et al., 2023), we focus our analysis on the impact of LLMs, rather than addressing machine learning or automation technologies more broadly. Additionally, we propose a novel method that employs LLMs, specifically GPT-4, to assess tasks for exposure and automation potential, thereby bolstering human scoring efforts. Subsequently, we aggregate our findings to occupations and industries, capturing the overall potential exposure in the contemporary U.S. labor market.

3 Methods and Data Collection

3.1 Data on Activities and Tasks Performed by Occupation in the US

We use the O*NET 27.2 database (O*NET, 2023), which contains information on 1,016 occupations, including their respective Detailed Work Activities (DWAs) and tasks. A DWA is a comprehensive action that is part of completing task, such as "Study scripts to determine project requirements." A task, on the other hand, is an occupation-specific unit of work that may be associated with zero, one, or multiple DWAs. We offer a sample of tasks and DWAs in Table 1. The two datasets we use consist of:

- 19,265 tasks, consisting of a "task description" and a corresponding occupation, and
- 2,087 DWAs, where most DWAs are connected to one or more tasks, and tasks may be associated with one or more DWAs, though some tasks lack any associated DWAs.

3.2 Data on Wages, Employment, and Demographics

We obtain employment and wage data from the 2020 and 2021 Occupational Employment series provided by the Bureau of Labor Statistics. This dataset encompasses occupational titles, the number of workers in each occupation, and occupation-level employment projections for 2031, typical education required for entry in an occupation and on-the-job training required to attain competency in an occupation (BLS, 2022). We use the BLS-recommended crosswalk to O*NET (BLS, 2023b) to link the O*NET task and DWA dataset and the BLS Labor Force Demographics (BLS, 2023a), which is derived from the Current Population Survey (CPS). Both of these data sources are collected by the U.S. government and primarily capture workers who are not self-employed, are documented, and are working in the so-called formal economy.

3.3 Exposure

We present our results based on an exposure rubric, in which we define **exposure** as a measure of whether access to an LLM or LLM-powered system would reduce the time required for a human to perform a specific

5 Validation of Measures

5.1 Comparison to Earlier Efforts

This paper aims to build on a number of previous empirical studies examining the occupational exposure to advances in AI and/or automation. Previous studies have used a variety of methods, including:

- Using occupational taxonomies like O*NET to characterize which occupations have routine vs. non-routine and manual vs. cognitive task content (Autor et al., 2003; Acemoglu and Autor, 2011a).
- Mapping text descriptions of tasks to descriptions of technological advances in patents. (Kogan et al., 2021; Webb, 2020)
- Linking capabilities of AI systems to occupational abilities and aggregating exposure estimates to the occupations where those abilities are required. (Felten et al., 2018, 2023)
- Mapping the results of AI task benchmark evaluations (ImageNet, Robocup, etc.) to 59 worker tasks through a set of 14 cognitive abilities drawn from the cognitive science literature. (Tolan et al., 2021)
- Expert labeling of automation potential for a set of O*NET occupations where experts had high confidence, combined with a probabilistic classifier to estimate automation potential for the remainder of O*NET occupations. (Frey and Osborne, 2017)
- Developing a rubric for evaluating the "suitability for machine learning" (SML) of activities that workers are completing in the economy (Brynjolfsson and Mitchell, 2017; Brynjolfsson et al., 2018, 2023).

We provide a set of summary statistics on many of these prior efforts in Table 8.

This paper's methodology primarily builds upon the SML approach by developing a rubric to evaluate the overlap between LLM capabilities and worker tasks as reported in the O*NET database. Table 9 presents the results of OLS regressions of our new LLM exposure measurements on occupation-level exposure measures from (Felten et al., 2018) ("AI Occupational Exposure Score" in the table), (Frey and Osborne, 2017) (Frey & Osborne Automation), scores from all three technologies in (Webb, 2020), normalized routine manual and cognitive scores from (Acemoglu and Autor, 2011a), and (Brynjolfsson et al., 2018, 2023) (SML). We also use annualized occupational salaries from the most recent BLS Occupational Employment Survey as a control. There are four separate output variables representing new scores in this paper that are predicted by earlier efforts.

GPT-4 Exposure Rating 1 corresponds to our overall exposure rubric as evaluated by GPT-4, where full exposure potential is coded as 1, no exposure potential is coded as 0, and partial exposure (E2 in our labeling scheme) is coded as 0.5. GPT-4 Exposure Rating 2 is scored similarly for overall exposure, but with a slightly different prompt. The results are very similar across the two prompts. Human Exposure Rating represents the same rubric as in GPT-4 Exposure Rating 1 but is scored by humans, as discussed in an earlier section of the paper. These results correspond to the β set of statistics presented above, with supplemental tasks having half the weight of core tasks within an occupation. These weights sum to one (core/supplemental distinctions are determined by O*NET).

The results across each type of measurement are consistent. We find generally positive and statistically significant correlations between our LLM exposure measures and previous measurements targeting software and AI. Encouragingly, the SML exposure scores by occupation show significant and positive associations with the exposure scores we develop in this paper, demonstrating a level of cohesion between the two studies with similar approaches. The Webb software and AI patent-based measures, SML, and normalized (demeaned